Guidance-Coupled Evaluation of Bearing-Only Optical Navigation in Cislunar Space

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Abstract

This work evaluates bearing-only lunar optical navigation through the midcourse correction it enables, rather than through estimator statistics alone. A simulated spacecraft observes the Moon with a calibrated pinhole camera; idealized body-center pixel measurements are converted to tangent-plane line-of-sight residuals, fused with a CR3BP iterated extended Kalman filter (IEKF), and passed to STM-based single-impulse targeting. The numerical experiments test whether that minimal bearing stream becomes guidance-useful when target visibility is actively maintained. In the baseline 1000-trial CR3BP Monte Carlo with one-pixel measurement noise, the estimated-state correction achieved a median terminal miss of 70.6 km and 99.1% success for a 10^{-3} LU tolerance, while fixed pointing preserved target visibility for only 24.6% of the active-tracking arc versus 99.7% with estimate-driven pointing. Stress tests show that innovation consistency alone is insufficient: NIS can remain nominal while NEES, timing sensitivity, or terminal miss reveal overconfidence and model mismatch. The result is a simulation baseline for a candidate Moon-center bearing architecture, not a claim of mature image processing or flight validation.

Nomenclature

$\mathbf{x}$	spacecraft state, $[\mathbf{r}^T, \mathbf{v}^T]^T \in \mathbb{R}^6$
μ	CR3BP mass parameter for the Earth and Moon system
$\Phi(t, t_0)$	state transition matrix (STM)
$\mathbf{P}$	state-estimation covariance
$\mathbf{Q}_d$	discrete process-noise covariance
q_a	white-acceleration process-noise spectral density
$\hat{\mathbf{u}}$	line-of-sight (LOS) unit vector
$\mathbf{E}$	tangent-plane projection basis for bearing residuals
$\mathbf{H}$	measurement sensitivity matrix
$\mathbf{R}$	bearing measurement-noise covariance
$\mathbf{K}$	Kalman gain
t_c	correction-burn time
NIS	normalized innovation squared
NEES	normalized estimation error squared
LU, TU	CR3BP nondimensional length and time units

1 Introduction

Future cislunar missions will operate in a regime where ground tracking is valuable but not always sufficient as the only navigation source. Line-of-sight access can be intermittent, antenna resources are shared, and the geometry of the Earth and Moon system can make updates sparse near mission-design-sensitive phases. A camera is attractive because it is passive, lightweight, and already present on many spacecraft for inspection, landing, or science. The limitation is equally important: a single image of a body gives direction, not range. Bearing-only optical navigation therefore succeeds only if the dynamics and changing observation geometry supply enough information to recover the missing depth state.

The purpose of this study is to test that idea in a closed-loop cislunar setting. The contribution is a guidance-coupled evaluation of a minimal bearing-only optical navigation architecture, rather than a new filtering algorithm. The core hypothesis is:

> Bearing measurements, when fused with nonlinear cislunar dynamics and protected by active pointing, can produce state estimates accurate enough to compute near-benchmark correction maneuvers.

This work makes three contributions. First, it formulates a Moon-center bearing-only optical navigation pipeline in which pixel measurements are converted to tangent-plane line-of-sight residuals and fused with a CR3BP IEKF. Second, it evaluates the estimator through an STM-based midcourse correction, using terminal miss and burn deviation as primary guidance metrics. Third, it shows through Monte Carlo, active-pointing, and model-mismatch studies that innovation consistency alone is insufficient; visibility, NEES, and terminal guidance performance must be assessed jointly.

The resulting analysis is a minimal-information baseline for cislunar optical navigation: one target, one Moon-center bearing stream, no radiometric range, no landmark updates, and no multi-body triangulation. The question is whether nonlinear dynamics and actively maintained measurement availability can turn that sparse information stream into a correction maneuver that remains close to the perfect-information benchmark.

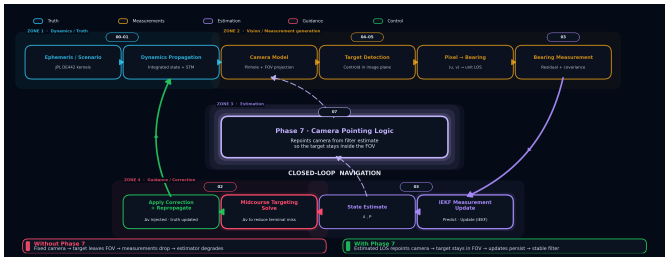

Figure 1: Closed-loop simulation architecture used in this study. The truth model generates the cislunar trajectory, the camera model generates pixel observations, the IEKF produces the navigation state, targeting computes the midcourse burn, and active pointing closes the measurement-availability loop.

2 Related Work

Optical navigation has been widely studied in both planetary and cislunar contexts, but the measurement products differ substantially across the literature. Horizon- or limb-based methods use the apparent conic of an extended body to estimate position, attitude, or both; Christian's tutorial gives a complete treatment of that class of algorithms [5]. Cislunar navigation studies have also considered natural and artificial optical targets, lunar landmarks, and horizon-based measurements in multi-sensor or higher-fidelity settings [6, 7]. Deep-space navigation often combines optical plane-of-sky information with radiometric tracking, where range and Doppler directly improve line-of-sight observability [8]. Recent flight demonstrations such as LONEStar further show the value of optical triangulation with multiple planetary targets [9].

This work intentionally removes several of those information sources. The simulated sensor does not fit a lunar limb, track a known landmark in the filter, use radiometric range/Doppler, or triangulate against multiple planets. It uses a Moon-center bearing derived from idealized body-center pixel measurements, so each accepted frame contributes only two angular constraints. While a direct numerical comparison is not strictly one-to-one because the measurement models differ, the results provide a minimal-information baseline for bearing-only sensing.

This work isolates a lower-information case rather than optical navigation in general. Prior work has demonstrated richer optical observables, landmark navigation, horizon-based navigation, or optical/radiometric fusion. This study focuses on a single Moon-center bearing stream and evaluates whether the resulting state estimate remains useful for midcourse correction. The comparison metric is therefore not image-processing accuracy alone, but terminal miss, burn deviation from a perfect-information correction, and estimator consistency.

3 Mission Scenario

The nominal scenario is a southern halo/near-rectilinear halo orbit (NRHO)-class arc near the Earth and Moon

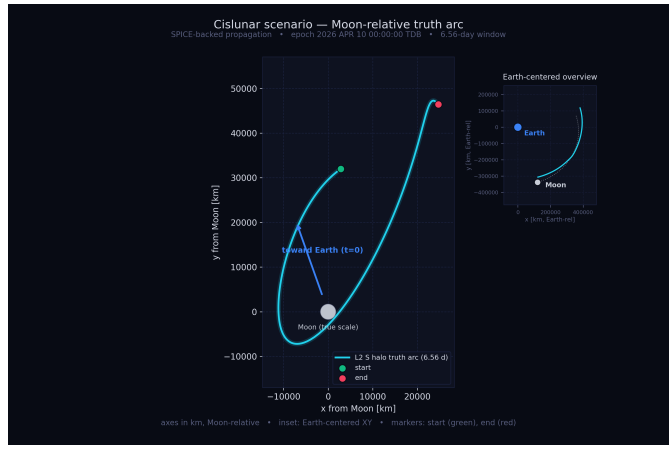

Figure 2: SPICE-backed scenario visualization for the representative 6.56 day cislunar tracking arc. The main panel shows the Moon-relative truth arc with the Moon drawn to scale; the inset shows the Earth-centered geometry. These physical units anchor the nondimensional CR3BP estimation and targeting experiments.

L_2 region, seeded from the JPL periodic-orbits catalog and visualized with SPICE-backed ephemeris data. The catalog record used for the study identifies the Earth and Moon system with

$$\mu = 1.215058560962404 \times 10^{-2},$$

$$1\,\mathrm{LU} = 389703.2648\ \mathrm{km},$$

and

$$1\,\mathrm{TU} = 382981.2891\ \mathrm{s} = 4.4327\ \mathrm{days}.$$

Most navigation and targeting quantities are expressed in nondimensional CR3BP units. The final Monte Carlo tolerance of 10^{-3} LU therefore corresponds to approximately 390 km. The representative arc spans 6.56 days and includes an insertion/tracking phase followed by a midcourse correction event.

The baseline estimation and targeting experiments use a fixed nominal trajectory, a fixed measurement schedule, and a correction time $t_c = 2.0$ in the nondimensional simulation used for the Monte Carlo products. The active-pointing comparison uses 301 candidate measurement epochs over the representative arc. In all cases, a measurement update is only applied when the Moon is visible in the camera field of view and passes the measurement gate.

4 Problem Statement

Given a spacecraft state $\mathbf{x} = [\mathbf{r}^T, \mathbf{v}^T]^T$ evolving under Earth–Moon CR3BP dynamics and a sequence of noisy Moon-center bearing measurements, estimate the spacecraft state at correction time t_c and compute a single impulsive correction that minimizes terminal position error at t_f. Each accepted image provides only two angular

constraints; range to the Moon and all velocity components must be inferred through the measurement history and dynamics rather than measured directly.

The estimator objective is to produce $\hat{\mathbf{x}}(t_c)$ and covariance $\mathbf{P}(t_c)$ that are accurate enough and statistically honest enough for guidance. The guidance objective is to compute an estimated-state burn $\Delta\mathbf{v}_E$ whose terminal miss and burn vector are close to the perfect-information burn $\Delta\mathbf{v}_P$ computed from the true state. Performance is evaluated by terminal miss, ΔV inflation relative to the perfect-information correction, NIS, NEES, and visibility fraction.

5 Dynamics and Propagation

5.1 CR3BP Model

The primary filter and targeting experiments use the CR3BP for the Earth and Moon system in the synodic frame. The primaries are fixed at

$$\mathbf{r}_E = (-\mu, 0, 0)^T, \qquad \mathbf{r}_M = (1-\mu, 0, 0)^T.$$

For spacecraft position (x, y, z),

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \tag{1}$$

$$r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}, \tag{2}$$

and the pseudo-potential is

$$\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}. \tag{3}$$

The implemented equations of motion are

$$\dot{x} = v_x, \qquad \dot{y} = v_y, \qquad \dot{z} = v_z, \tag{4}$$

$$\dot{v}_x = 2v_y + \Omega_x, \qquad \dot{v}_y = -2v_x + \Omega_y, \qquad \dot{v}_z = \Omega_z. \tag{5}$$

The same `CR3BP` class also computes the Jacobi constant, the Hessian of Ω, the linearized dynamics matrix $\mathbf{A}$, and the collinear/triangular Lagrange points. This gives the estimator both a nonlinear propagation model and the variational dynamics needed for covariance transport.

5.2 State Transition Matrix

Covariance propagation uses the STM,

$$\Phi(t, t_0) = \frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}(t_0)}.$$

The state and STM are propagated together through

$$\dot{\Phi}(t, t_0) = \mathbf{A}(t)\Phi(t, t_0), \qquad \Phi(t_0, t_0) = \mathbf{I}_6. \tag{6}$$

In implementation, the state and column-major STM are packed into a 42-element vector. The filter step integrates this combined system with adaptive tolerances, unpacks the terminal state and STM, and updates covariance as

$$\bar{\mathbf{P}}_{k+1} = \Phi_k \mathbf{P}_k^+ \Phi_k^T + \mathbf{Q}_{d,k}. \tag{7}$$

Small numerical asymmetries are removed by symmetrizing $\mathbf{P}$. A positive-definiteness guard distinguishes harmless roundoff from true divergence.

5.3 Process Noise

The process-noise model is discrete white acceleration on the six-state vector:

$$\mathbf{Q}_d(\Delta t) = q_a \begin{bmatrix} \frac{\Delta t^3}{3}\mathbf{I}_3 & \frac{\Delta t^2}{2}\mathbf{I}_3 \\ \frac{\Delta t^2}{2}\mathbf{I}_3 & \Delta t \mathbf{I}_3 \end{bmatrix}. \tag{8}$$

This model is intentionally simple: it represents unmodeled accelerations, numerical/model mismatch, and the covariance inflation required to keep the filter honest. The final tuning study shows why this matters. Very small q_a values can produce accurate-looking point estimates while understating uncertainty, which appears as excessive NEES.

6 Optical Measurement Model

6.1 Pixel to Bearing

The camera is modeled with pinhole intrinsics

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}.$$

For a pixel detection $(u_{\text{px}}, v_{\text{px}})$, the normalized camera-frame ray is

$$\hat{\mathbf{u}}_C = \frac{[(u_{\text{px}} - c_x)/f_x,\ (v_{\text{px}} - c_y)/f_y,\ 1]^T}{\|[(u_{\text{px}} - c_x)/f_x,\ (v_{\text{px}} - c_y)/f_y,\ 1]^T\|}. \tag{9}$$

The ray is rotated into the navigation frame using the current camera attitude. A scalar pixel standard deviation is converted to angular noise using

$$\sigma_\theta \approx \frac{\sigma_{\text{px}}}{f_x}, \tag{10}$$

or a root-mean-square focal-length variant when $f_x \neq f_y$. The result is a noisy unit LOS vector and a two-dimensional angular covariance for the bearing residual.

The measurement simulator includes Gaussian pixel noise, optional random dropout, field-of-view clipping, behind-camera rejection, simple distortion hooks, and bounding-box body measurements. These effects are sufficient to connect camera geometry to the estimator without claiming a mature image-processing pipeline.

The center measurement should be interpreted as an idealized body-center bearing observable. The present study does not model illumination-dependent center-of-brightness bias, limb-fitting errors, exposure and thresholding effects, or star-field false detections. These effects are intentionally excluded so that the navigation and guidance consequences of a minimal bearing stream can be isolated.

6.2 Tangent-Plane Residual

Let $\mathbf{r}_{sc}$ be the spacecraft position and $\mathbf{r}_b$ be the observed body position, usually the Moon. The predicted LOS is

$$\hat{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{r}_b - \mathbf{r}_{sc}}{\|\mathbf{r}_b - \mathbf{r}_{sc}\|}. \tag{11}$$

Table 1: Camera and measurement parameters used for the baseline Monte Carlo.

Parameter	Value
Image size	640×480 px
Focal length (f_x, f_y)	$(400, 400)$ px
Principal point (c_x, c_y)	$(320, 240)$ px
Full FOV	$77.3° \times 61.9°$
Measurement cadence	0.02 TU (2.13 h)
Baseline pixel noise	1 px

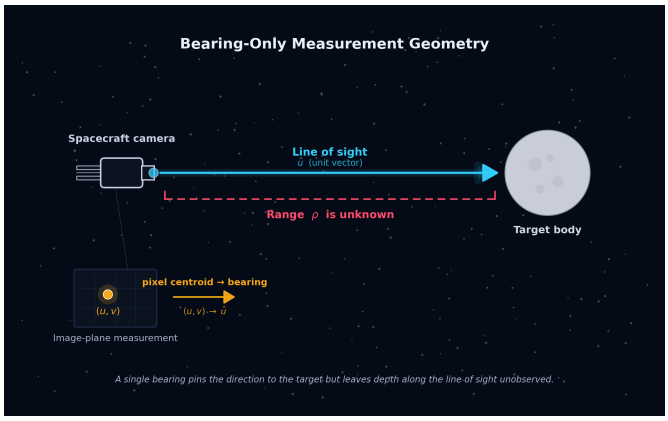

Figure 3: Bearing-only measurement geometry. The camera supplies an idealized body-center pixel location, which is converted to a line-of-sight unit vector; range along that ray is not directly measured and must be inferred from dynamics and repeated observations.

Using all three components of a unit vector would introduce a redundant constraint because $\|\hat{\mathbf{u}}\| = 1$. The implementation instead forms an orthonormal tangent basis $\mathbf{e}_1, \mathbf{e}_2$ perpendicular to the predicted LOS and stacks it as

$$\mathbf{E} = \begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \end{bmatrix}.$$

The two-dimensional innovation is

$$\mathbf{y}_k = \mathbf{E}\left(\hat{\mathbf{u}}_{\text{meas}} - \hat{\mathbf{u}}(\bar{\mathbf{x}}_k)\right). \tag{12}$$

For range $\rho = \|\mathbf{r}_b - \mathbf{r}_{sc}\|$, the LOS position sensitivity is

$$\frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{r}_{sc}} = -\frac{1}{\rho}\left(\mathbf{I}_3 - \hat{\mathbf{u}}\hat{\mathbf{u}}^T\right), \tag{13}$$

giving the measurement matrix

$$\mathbf{H} = \begin{bmatrix} \mathbf{E}\frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{r}_{sc}} & \mathbf{0}_{2\times 3} \end{bmatrix}, \qquad \mathbf{R} = \sigma_\theta^2 \mathbf{I}_2. \tag{14}$$

This is the mathematical center of the bearing-only estimator: one image contributes two angular constraints, and the range direction must be recovered over time through motion and dynamics.

7 Bearing-Only Observability

7.1 Instantaneous Rank

At a single image time, the tangent-plane bearing measurement is rank deficient by construction. The matrix in Eq. (14) has rank at most two because the measurement lives in the plane normal to the predicted LOS:

$$\operatorname{rank}(\mathbf{H}_k) \leq 2. \tag{15}$$

For position alone, this is the familiar "missing range" intuition: if the observed body location is known and the bearing direction is known, the spacecraft position lies somewhere on a line. For the full six-state problem, the instantaneous nullspace is larger because the measurement has no direct velocity sensitivity. One accepted frame leaves the LOS/range position direction and all three velocity directions unconstrained at that instant. Velocity and range only become observable through the time history of CR3BP motion and the way later LOS measurements depend on earlier position and velocity through the STM.

Over an arc, the relevant local observability object is the stacked linearized measurement history

$$\mathcal{O}_N = \begin{bmatrix} \mathbf{H}_1\Phi(t_1, t_0) \\ \mathbf{H}_2\Phi(t_2, t_0) \\ \vdots \\ \mathbf{H}_N\Phi(t_N, t_0) \end{bmatrix}. \tag{16}$$

Equivalently, the diagnostic calculation accumulates the unweighted Gramian

$$\mathbf{W}_N = \sum_{k=1}^{N} \Phi(t_k, t_0)^T \mathbf{H}_k^T \mathbf{H}_k \Phi(t_k, t_0), \tag{17}$$

with a noise-weighted version obtained by inserting $\mathbf{R}_k^{-1}$ between $\mathbf{H}_k^T$ and $\mathbf{H}_k$. Full rank of $\mathbf{W}_N$ is a necessary local test, but its conditioning is the more practical metric: a full-rank Gramian can still contain a very weak direction that makes range or velocity slow to recover. In practice, these weak directions correspond primarily to range and to certain velocity components aligned with the line of sight.

Bearing-only navigation does not become observable because an individual image contains range information. It becomes observable because errors in range and velocity alter future positions, which alter future bearing measurements through the STM.

7.2 Geometry Loss Cases

The measurement history loses practical observability when the LOS direction changes too little, when the relative motion is nearly straight-line over the tracking window, or when the STM does not couple velocity errors into later angular errors strongly enough before the correction time. It also loses observability in practice when pointing or field-of-view limits remove measurements, even if the mathematical trajectory would otherwise be informative. This explains why early correction times and fixed pointing are fragile in the results: the filter either has not yet accumulated enough angular parallax or has been starved of updates.

Figure 4 summarizes the accumulated observability Gramian for the representative trajectory. The accumulated Gramian is locally full rank along the simulated trajectory, but the final eigenvalues span roughly seven orders of magnitude, from $\lambda_1 \approx 8.5$ to $\lambda_6 \approx 10^8$. The weakest mode is concentrated in the in-plane $\{x, y, v_x, v_y\}$ subspace, while the next weak mode is dominated by v_z. This is the analytical version of the qualitative statement that range becomes recoverable over time: changing LOS geometry and nonlinear dynamics gradually rotate angular information into the missing state directions, but not all recovered directions are equally strong.

8 Estimator

8.1 Iterated EKF

The IEKF prediction is the STM propagation described above. The measurement update uses the standard innovation covariance

$$\mathbf{S}_k = \mathbf{H}_k \bar{\mathbf{P}}_k \mathbf{H}_k^T + \mathbf{R}_k$$

and Kalman gain

$$\mathbf{K}_k = \bar{\mathbf{P}}_k \mathbf{H}_k^T \mathbf{S}_k^{-1}.$$

Because the bearing measurement is nonlinear in position, the correction is iterated. With prior state $\bar{\mathbf{x}}$ and iteration state $\mathbf{x}^{(i)}$, the implemented correction has the form

$$\mathbf{x}^{(i+1)} = \bar{\mathbf{x}} + \mathbf{K}^{(i)} \left[\mathbf{y}^{(i)} + \mathbf{H}^{(i)} \left(\mathbf{x}^{(i)} - \bar{\mathbf{x}} \right) \right]. \tag{18}$$

The covariance is then updated with the Joseph form,

$$\mathbf{P}^+ = (\mathbf{I} - \mathbf{KH})\bar{\mathbf{P}}(\mathbf{I} - \mathbf{KH})^T + \mathbf{KRK}^T, \tag{19}$$

which is more robust to numerical loss of symmetry and positive definiteness than the simplified update.

8.2 Statistical Health Metrics

The estimator is evaluated with both measurement-space and state-space consistency tests. The normalized innovation squared is

$$\mathrm{NIS}_k = \mathbf{y}_k^T \mathbf{S}_k^{-1} \mathbf{y}_k. \tag{20}$$

For the two-dimensional tangent-plane residual, NIS should be distributed like $\chi^2(2)$ under correct modeling. The normalized estimation error squared is

$$\mathrm{NEES}_k = (\hat{\mathbf{x}}_k - \mathbf{x}_k)^T \mathbf{P}_k^{-1} (\hat{\mathbf{x}}_k - \mathbf{x}_k), \tag{21}$$

which is compared to $\chi^2(6)$ for the six-state estimate. The diagnostics suite also plots innovation histories, visibility gating, position/velocity 3σ envelopes, image-plane detections, and hypothesis checks.

The χ^2 bands are used as diagnostic reference levels for consistency; trial-mean NIS/NEES values are not treated as exact independent-sample hypothesis tests because measurements along a trajectory are temporally correlated.

9 Guidance Coupling

The estimator is not evaluated solely by state error because the guidance product is a correction maneuver. A navigation error is important only to the extent that it projects into burn-sensitive directions at t_c. Therefore, this section evaluates the map from estimation error to terminal guidance error.

The guidance problem asks whether the estimated state is good enough to compute a useful midcourse correction. A single impulsive burn is applied at correction time t_c. For a candidate burn $\Delta\mathbf{v}$, the state at t_c is modified as

$$\mathbf{x}_c^+ = \begin{bmatrix} \mathbf{r}_c \\ \mathbf{v}_c + \Delta\mathbf{v} \end{bmatrix}.$$

The targeting routine propagates from t_c to t_f with the STM and solves for a burn that drives the terminal position toward a desired target position. Linearizing the terminal position with respect to the burn gives

$$\delta\mathbf{r}_f \approx \Phi_{rv}(t_f, t_c)\, \delta\mathbf{v}_c, \tag{22}$$

where Φ_{rv} is the upper-right 3×3 block of the STM from correction time to final time. Newton iterations update

$$\Delta\mathbf{v}_{j+1} = \Delta\mathbf{v}_j - \Phi_{rv}^{-1} \left(\mathbf{r}_{f,j} - \mathbf{r}_{\mathrm{target}} \right), \tag{23}$$

using a least-squares fallback if the block is ill-conditioned. Three cases are compared: no correction, perfect-information correction from the true state, and IEKF correction from the estimated state.

The central diagnostic is the map from navigation error to burn error. Figure 6 probes that map by perturbing the estimated correction-time state and recomputing the burn. The result explains why the observability eigenvectors in Fig. 4 matter: the weakest in-plane mode is both poorly observed and burn-coupled, while the weak out-of-plane v_z mode is much less damaging to this planar transfer. The guidance problem is therefore not controlled by observability rank alone; it is controlled by the alignment between weak navigation directions and terminal targeting sensitivity.

10 Active Camera Pointing

10.1 Policy

Fixed body pointing is a fragile assumption in cislunar optical navigation. If the Moon drifts out of frame, measurements disappear and the filter reverts to open-loop propagation in a strongly nonlinear dynamical environment. The active pointing policy therefore treats camera orientation as part of the navigation architecture. At each measurement epoch, the current state estimate and target position define a desired LOS,

$$\hat{\mathbf{b}}_{\mathrm{cmd}} = \frac{\mathbf{r}_b - \hat{\mathbf{r}}_{sc}}{\|\mathbf{r}_b - \hat{\mathbf{r}}_{sc}\|}.$$

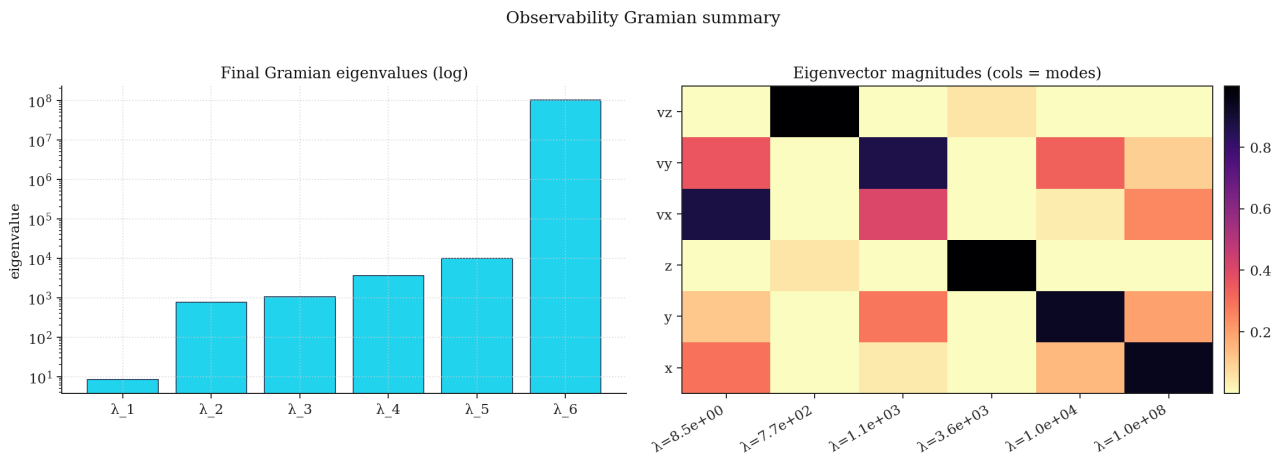

Figure 4: Final observability Gramian summary for the representative cislunar baseline. The eigenvalue spread confirms that bearing-only navigation is locally full rank but strongly conditioned by geometry. The eigenvector map identifies which state components dominate each weak mode.

Table 2: Fixed versus estimate-driven active pointing on the 301-step active-tracking experiment. Errors are in nondimensional CR3BP units.

Metric	Fixed	Active
Visibility fraction	0.2458	0.9967
Update fraction	0.2458	0.9967
Median NIS	1.17	1.32
RMS position error	5.44×10^{-1}	4.12×10^{-4}
Final position error	2.45	6.54×10^{-6}
RMS velocity error	5.84×10^{-1}	1.38×10^{-3}
Median pixel offset	88.8 px	1.20 px

The camera direction-cosine matrix is rebuilt so that the camera boresight aligns with this commanded LOS while an "up" hint resolves roll. The simulator then projects the true target into that camera frame. This closes the loop:

$$\hat{\mathbf{x}} \to \text{camera attitude} \to \text{pixels} \to \text{bearing} \to \hat{\mathbf{x}}.$$

10.2 Effect on Observability

Table 2 shows the effect. The fixed camera only preserves visibility for 24.6% of the 301 measurement epochs; active pointing preserves 99.7%. The fixed case eventually diverges because the filter stops receiving useful measurements, while active pointing holds the target near the principal point and keeps the estimator stable.

11 Monte Carlo Results

11.1 Baseline Study

The baseline Monte Carlo uses 1000 trials, $\sigma_{\text{px}} = 1$ pixel, no random dropout, estimate-driven camera pointing, and tuned process noise selected from the fine sweep. Each trial samples the initial truth-injection perturbation, initial estimation perturbation, and pixel measurement noise

Table 3: Baseline Monte Carlo, $n = 1000$, $\sigma_{\text{px}} = 1$ pixel, estimate-tracking, $t_c = 2.0$. Relative ΔV values are converted to percent for readability.

Metric	Median	5th pct.	95th pct.
Terminal miss [LU]	1.81×10^{-4}	3.79×10^{-5}	5.64×10^{-4}
Terminal miss [km]	70.6	14.8	219.6
$\lVert \Delta\mathbf{v}_E - \Delta\mathbf{v}_P \rVert$	2.92×10^{-4}	6.97×10^{-5}	8.19×10^{-4}
$\lvert \Delta V_E \rvert - \lvert \Delta V_P \rvert$	2.24×10^{-5}	-3.55×10^{-4}	4.58×10^{-4}
Relative ΔV inflation	+1.19%	−31.9%	+51.4%
Position error at t_c [LU]	5.28×10^{-5}	1.83×10^{-5}	1.23×10^{-4}
Trial-mean NIS	1.94	1.63	2.28
Trial-mean NEES	4.34	2.25	8.52

using deterministic random seeds. The nominal trajectory, measurement schedule, camera model, and correction time are fixed across trials unless otherwise stated. Pixel noise is independent across accepted frames, and measurement updates are applied only when the target is inside the field of view and passes gating. Terminal miss is computed after applying the estimated-state correction and propagating to t_f. A trial is counted as successful if the post-correction terminal miss is below 10^{-3} LU.

Initial truth-injection and estimate perturbations are sampled independently per component from zero-mean Gaussians with $\sigma_r = 10^{-4}$ LU and $\sigma_v = 10^{-4}$ LU/TU. The IEKF is initialized with $\mathbf{P}_0 = \text{diag}(10^{-6}\mathbf{I}_3, 10^{-7}\mathbf{I}_3)$, where the first block is position covariance in LU^2 and the second block is velocity covariance in (LU/TU)^2.

Table 3 summarizes the main result. The IEKF correction is not identical to the perfect-information burn, but the navigation error is small enough that the closed-loop correction usually meets tolerance. The median terminal miss is 1.81×10^{-4} LU, or approximately 70.6 km. The 95th percentile is 5.64×10^{-4} LU, or approximately 220 km, still below the specified tolerance. The uncorrected median miss is 1.19×10^{-3} LU, so the estimated-state correction improves the typical terminal error by a factor of roughly 6.6.

Negative relative ΔV inflation indicates that the

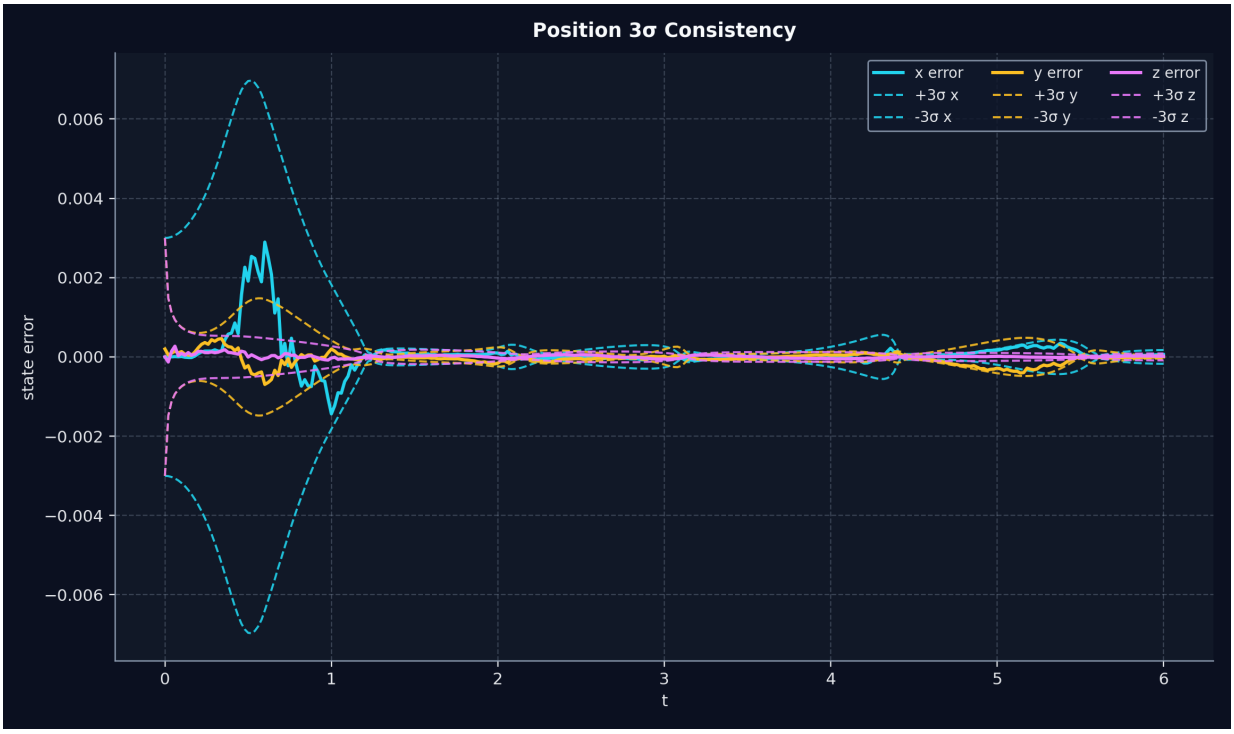

Figure 5: Representative position-error and 3σ consistency check for the estimate-tracking baseline. The component errors remain within the estimated covariance envelope after the early transient, supporting the NEES/NIS consistency claims.

estimated-state correction has smaller magnitude than the perfect-information correction. It does not imply superior targeting performance; the terminal miss remains the primary measure of correction quality.

The distribution products are as important as the summary table because they show whether errors are systematic or trial-dependent. Figure 8 consolidates the final baseline into four views: the terminal-miss ECDF, the relative burn-cost distribution, the map from correction-time position error to terminal miss, and the NIS/NEES consistency distributions. The positive trend in the scatter panel is expected: state error at t_c maps through the targeting STM into the burn and final miss. The spread around that trend reflects trajectory geometry and the conditioning of Φ_{rv}, so a scalar position error alone is not a complete predictor of guidance quality.

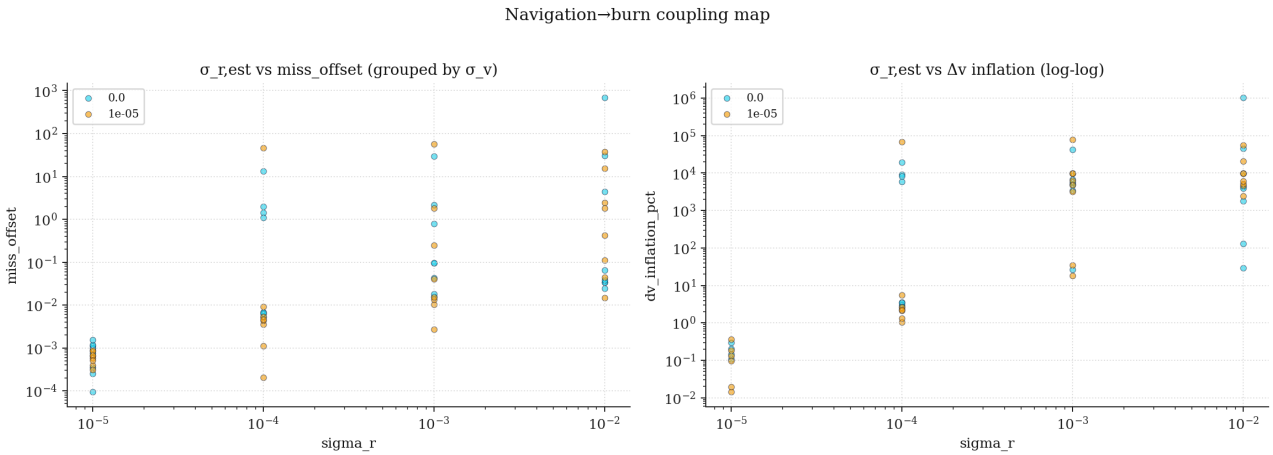

Figure 6: Navigation-error to burn-error coupling map for the representative cislunar baseline. Terminal miss and relative ΔV inflation increase rapidly with the correction-time position-error scale, showing why navigation performance should be evaluated at the guidance output.

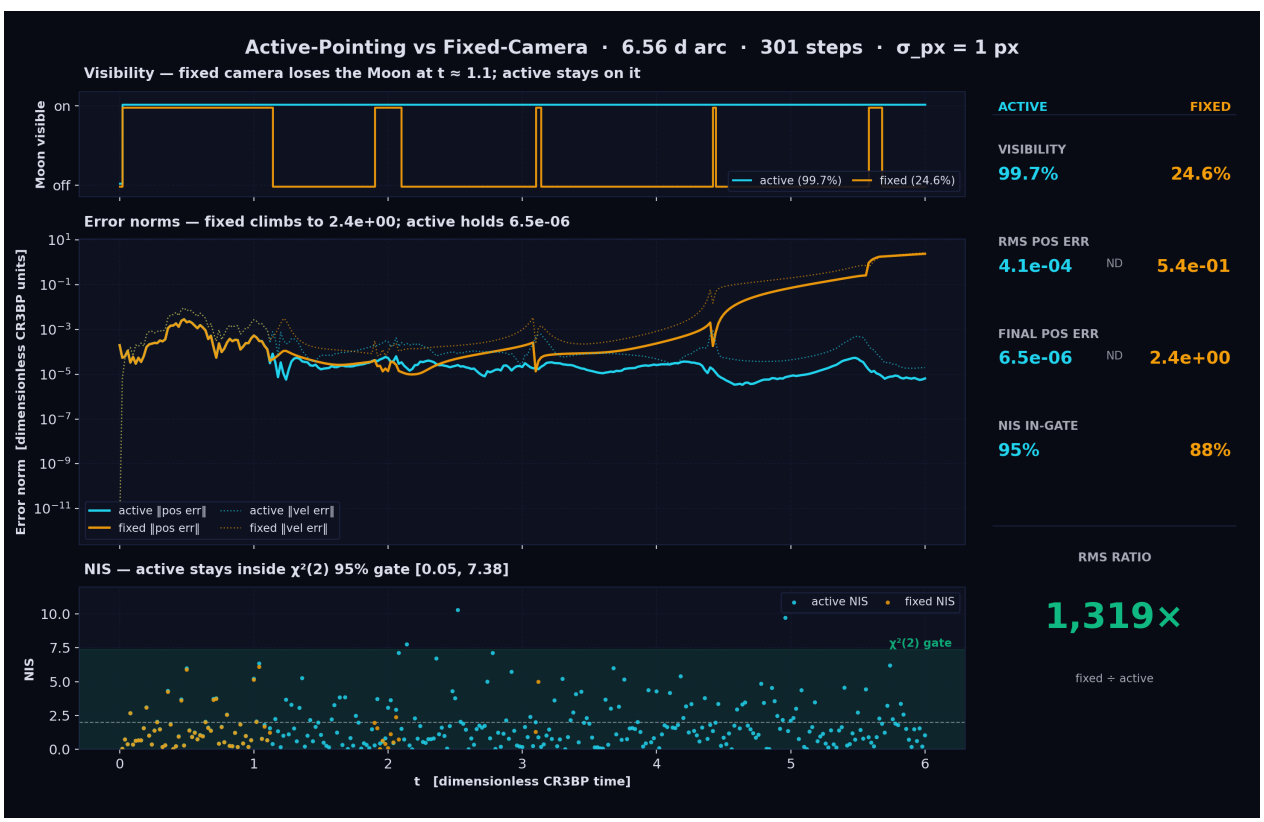

Figure 7: Fixed and active pointing comparison. Fixed pointing loses target visibility and produces estimator divergence, while estimate-driven pointing maintains measurement availability and bounded estimation error.

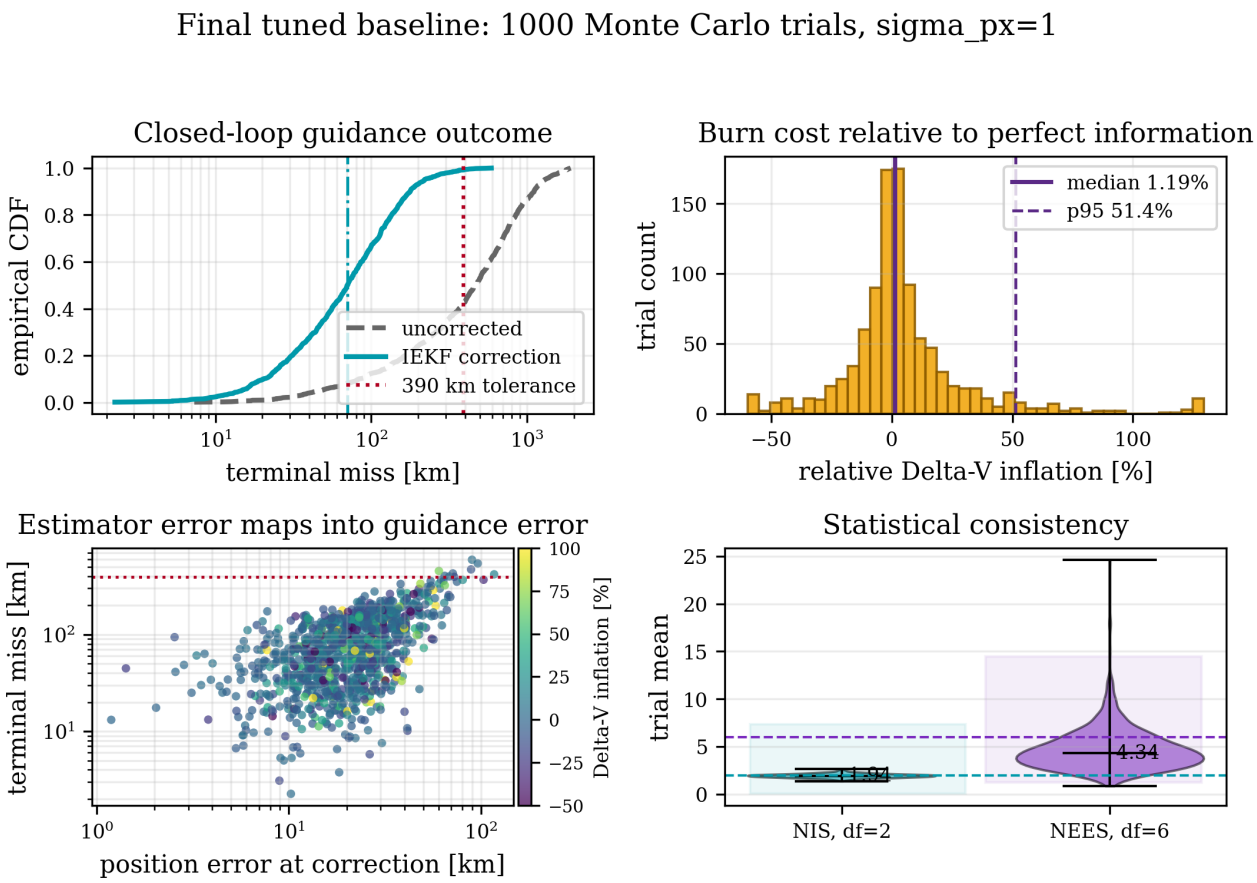

Figure 8: Baseline Monte Carlo summary for 1000 trials. The IEKF-fed correction shifts the terminal-miss distribution below the 10^{-3} LU tolerance, retains a small median ΔV penalty, and remains statistically credible in both measurement space and state space.

11.2 High-Noise Stress Case

A high-noise stress case with $\sigma_{\mathrm{px}} = 5$ pixels keeps the same nominal architecture but widens the measurement uncertainty. The median terminal miss increases to 1.72×10^{-3} LU and the success rate drops to 29.5%. The 95th percentile miss is 5.27×10^{-3} LU. This behavior is expected: bearing-only navigation has no direct range measurement, so degraded angular accuracy enters both state reconstruction and burn targeting. The result is useful because it identifies where the one-target, body-center architecture becomes fragile.

11.3 Process-Noise Tuning

The process-noise study was performed in two stages. A coarse sweep exposed the main failure mode: too little process noise made the filter overconfident, with NEES far above the $\chi^2(6)$ range. A later fine sweep ran 1000 trials at each of 11 candidate q_a values in the final operating regime. Table 4 shows that $q_a = 10^{-9}$ preserved the 99.1% pass rate while moving the median NEES and in-band fraction into a statistically credible range.

12 Sensitivity Studies

The sensitivity studies vary pixel noise and correction time while holding the rest of the pipeline fixed. Section 13 adds a wider set of sensitivity sweeps (terminal tolerance, attitude uncertainty, parallax-versus-range, and a synthetic-landmark variant) at the production seed count of $n = 1000$. The pixel-noise and t_c sensitivity here uses an earlier $n = 80$ deterministic-seed configuration with active pointing and $q_a = 10^{-9}$. Pixel noise remains a manageable driver through $\sigma_{px} = 3$: the median terminal miss stays below the 10^{-3} LU tolerance, rising from 1.22×10^{-4} LU at $\sigma_{px} = 0.5$ to 6.95×10^{-4} LU at $\sigma_{px} = 3$. At $\sigma_{px} = 5$,

Table 4: Fine process-noise sweep, 1000 trials per point. "Band" is the fraction of trial-mean NEES values inside the diagnostic $\chi^2(6)$ 95% interval.

q_a	Pass	Miss med.	NEES med.	Band
1.0×10^{-12}	0.991	1.58×10^{-4}	5.77	0.893
3.16×10^{-12}	0.991	1.57×10^{-4}	5.74	0.899
1.0×10^{-11}	0.991	1.58×10^{-4}	5.71	0.905
3.16×10^{-11}	0.991	1.58×10^{-4}	5.57	0.925
1.0×10^{-10}	0.991	1.57×10^{-4}	5.22	0.944
3.16×10^{-10}	0.991	1.67×10^{-4}	4.82	0.977
1.0×10^{-9}	0.991	1.81×10^{-4}	4.34	0.994
3.16×10^{-9}	0.989	2.33×10^{-4}	3.92	0.998
1.0×10^{-8}	0.970	2.90×10^{-4}	3.62	0.996
3.16×10^{-8}	0.942	3.37×10^{-4}	3.30	0.995
1.0×10^{-7}	0.904	3.75×10^{-4}	2.97	0.995

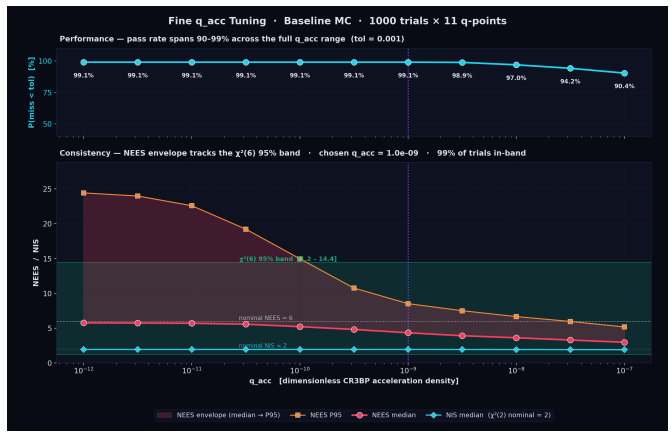

Figure 9: Fine q_a tuning summary. The selected region keeps terminal miss performance close to the baseline while making the estimated covariance statistically credible.

the median reaches 1.73×10^{-3} LU, and the upper tail is well outside tolerance. Correction time shows a separate geometric effect at $\sigma_{px} = 1.5$: the median miss drops from 1.32×10^{-3} LU at $t_c = 0.8$ to 1.55×10^{-4} LU at $t_c = 3.0$, although the $t_c = 1.2$ and 1.6 points retain rare high-miss outliers before the geometry has fully matured.

13 Realism and Sensitivity Extensions

The baseline study assumes ideal pointing, an optimistic terminal-miss tolerance, and a single Moon-center bearing source. This section relaxes those assumptions one at a time on top of the same 1000-trial pipeline. The Monte Carlo seeds are deterministic so per-trial deltas are meaningful. Throughout this section, "parallax" refers to the cumulative line-of-sight angular change accumulated step-to-step over the observation arc, not the endpoint-to-endpoint angle (these differ on multi-revolution arcs and the cumulative form is the geometrically correct measure of bearing-information path length). All NIS aggregations follow the bookkeeping convention from the analysis-driver documentation: `nis_mean` is the Moon-center NIS, `nis_mean_all` pools Moon-center and accepted-landmark

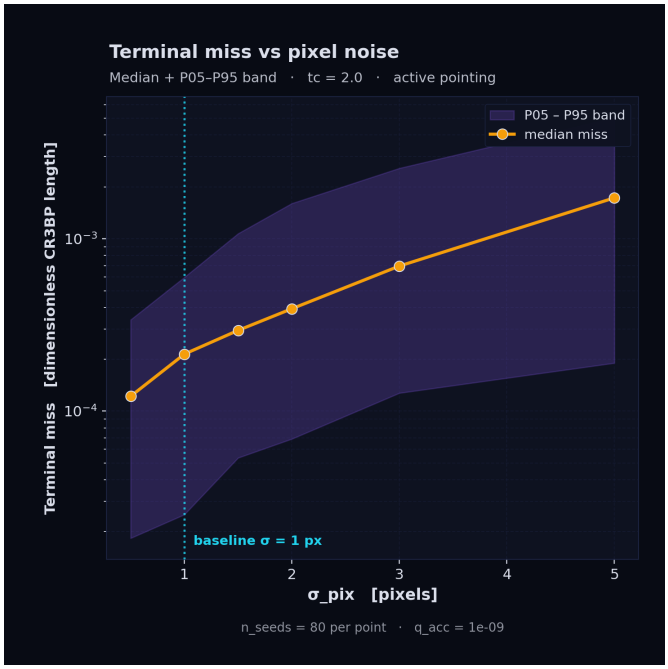

Figure 10: Terminal miss sensitivity to pixel measurement noise. Each point aggregates 80 deterministic seeds; the logarithmic axis makes both the median trend and high-miss tail visible.

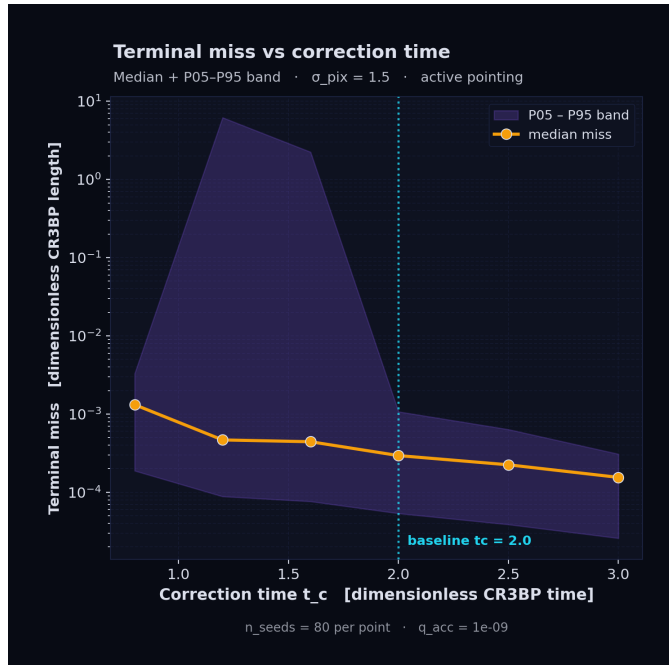

Figure 11: Terminal miss sensitivity to correction time using 80 deterministic seeds at $\sigma_{px} = 1.5$. The result highlights that guidance performance depends on both estimator maturity and remaining maneuver authority.

updates, and `nis_mean_landmarks` restricts to landmark accepts; mixed-source configurations report the pooled column.

13.1 Terminal-Tolerance Reality Check

The baseline tolerance is 10^{-3} LU ≈ 390 km, which is generous relative to mission-relevant accuracy requirements. Sweeping the success threshold over $[1, 1000]$ km on the existing $n = 1000$ baseline (Figure 12) shows where the headline pass rate degrades. At 10^{-3} LU the CR3BP-truth pass rate is 99.2% and the SPICE-truth pass rate is 85.3%. At a tighter 10^{-4} LU ≈ 39 km tolerance the pass rates fall to 32.9% (CR3BP) and 11.2% (SPICE). At 25 km the rates are 18.9% and 6.8%. The tolerance reality check is therefore that the baseline architecture meets the loose threshold comfortably but is not yet a precision-arrival tool: the median miss is well-controlled, but the trial-to-trial spread leaves a substantial fraction of trajectories outside the tighter requirement, and the SPICE truth case sits on a measurably worse part of the curve.

13.2 Attitude Uncertainty

The baseline assumes ideal pointing. To stress-test that assumption, a small zero-mean rotational noise is applied to the camera attitude at every measurement step. Specifically, an axis-angle vector $\theta \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{att}}^2 \mathbf{I}_3)$ is sampled, the small-angle rotation $\mathbf{R}_{\text{perturb}} = \exp([\theta]_\times)$ is composed with the commanded camera DCM (truth side only), and the filter back-projects bearings through the unperturbed commanded DCM. Attitude noise therefore enters the filter as un-modeled measurement-side error rather than as a known noise source the filter could absorb into $\mathbf{R}_k$. Sweeping $\sigma_{\text{att}} \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1.0\}$ degrees at $n = 1000$ trials per point produces Figure 13 and Table 5. The behavior is graceful degradation followed by collapse. The median terminal miss is statistically indistinguishable from the no-noise baseline through $\sigma_{\text{att}} = 0.03$ deg (60.8–63.6 km), inflates to 86.8 km at $\sigma_{\text{att}} = 0.1$ deg, and then climbs steeply to 227.9 km at 0.3 deg and 1043.6 km at 1.0 deg. Filter consistency, measured by Moon-center NIS and the fraction of trial-mean NEES values inside the diagnostic $\chi^2(6)$ 95% interval, tracks the same regime change: NEES is in-band for at least 89% of trials up to $\sigma_{\text{att}} = 0.03$ deg, drops to 76% at 0.1 deg, and collapses below 2% at 0.3 deg. The corresponding NIS column is `nis_mean` since this configuration only uses the Moon-center bearing.

Table 5: Attitude-noise sweep, $n = 1000$ trials per σ_{att}. The third and fourth columns are the median and 95th-percentile terminal miss in kilometres, the fifth is median Moon-center NIS, and the last is the fraction of trial-mean NEES values inside the diagnostic $\chi^2(6)$ 95% interval.

σ_{att} [deg]	n	miss median [km]	miss p95 [km]	NIS med.	NEES band frac.
0.000	1000	60.80	225.16	1.94	0.893
0.001	1000	62.48	230.62	1.95	0.910
0.003	1000	62.88	228.27	1.95	0.909
0.010	1000	62.35	229.41	1.96	0.902
0.030	1000	63.61	240.87	2.04	0.893
0.100	1000	86.81	317.51	2.89	0.760
0.300	1000	227.88	824.59	10.50	0.019
1.000	1000	1043.62	3115.48	97.88	0.000

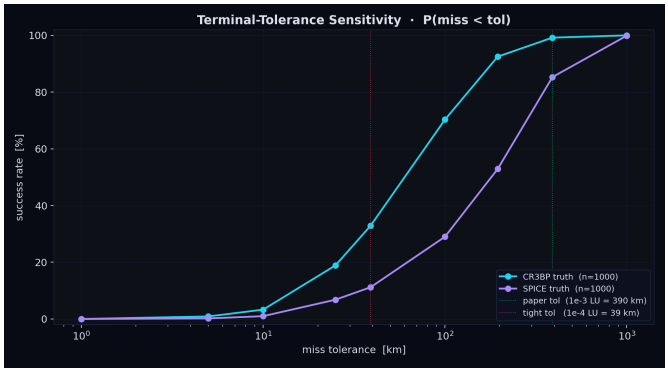

Figure 12: Terminal-miss success rate as a function of tolerance, taken from the same $n = 1000$ CR3BP-truth and SPICE-point-mass-truth baseline runs. Vertical guides mark the paper threshold (10^{-3} LU = 390 km) and the tighter 10^{-4} LU = 39 km point used as the reality-check level.

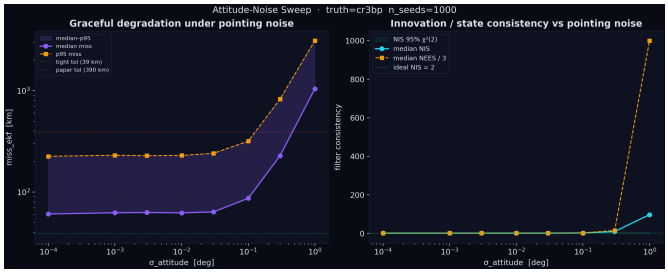

Figure 13: Attitude-noise sensitivity sweep, $n = 1000$ trials per σ_{att} point. The left panel shows median and 95th-percentile terminal miss; the right panel shows the median Moon-center NIS and median NEES/3. The performance plateau extends to roughly $\sigma_{\text{att}} = 0.1$ deg before the filter becomes inconsistent and miss climbs steeply.

13.3 Cumulative-Parallax Range Observability

A bearing-only filter cannot observe range directly. Range becomes observable through the cumulative angular change of the Moon line-of-sight relative to the trajectory: the longer the spacecraft drifts in directions that rotate the LOS, the more parallax the filter accumulates, and the better the recovered range. Sweeping the correction time t_c over $\{0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0\}$ at $n = 1000$ trials per t_c yields one (parallax, range error) point per trial. Figure 14 plots range error at t_c as a function of cumulative parallax in degrees, with a power-law fit. The fit on the production data gives an exponent of -0.57 against cumulative parallax and -0.71 against the endpoint-to-endpoint quantity; the cumulative-parallax fit is the geometrically appropriate one because the trajectory is multi-revolution. Both exponents are negative and order-unity, consistent with the bearing-only intuition that range error decays with accumulated LOS sweep, but they are noticeably weaker than a clean inverse law. The departure from a clean -1 slope is informative: the very-short-arc bins ($t_c = 0.25, 0.5$) are pre-convergence regimes where the filter has not yet had enough updates to make use of available parallax, and the long-arc bins flatten at the floor set by pixel noise and CR3BP integration error. The figure therefore provides a physics-backed answer to the standard bearing-only question of why range eventually becomes observable, while indicating that range information is bounded both below (insufficient arc) and above (measurement and dynamics noise floor) rather than a single power law.

13.4 Landmark-Bearing Extension

A natural follow-on to the Moon-center bearing-only baseline is whether known surface features (craters, basin centers, named regions) can substitute for or augment the body-center bearing. The full image-recognition pipeline (frame rendering, crater detection, frame-to-frame tracking, catalog association, false-match rejection) is its own research scope; this paper isolates the navigation-geometry question by considering three increasingly realistic levels of landmark integration:

L1 *Synthetic known landmarks.* Six Moon-fixed unit-vector offsets along $\pm x, \pm y, \pm z$ in the body frame, scaled by one lunar radius. Geometric upper bound on the achievable improvement from spatial bearing diversity.

L2 *Catalog-backed landmark bearings.* Real selenographic latitude/longitude coordinates for six well-known nearside craters (Tycho, Copernicus, Aristarchus, Plato, Kepler, Grimaldi), converted to Moon-fixed Cartesian and used directly as bearing sources. Landmark identity is assumed known; image-recognition performance is not modeled. This level isolates the navigation-geometry question from computer-vision performance and is the largest scope addressed here.

L3 *Image-based identification with catalog association.* Render or ingest images, detect craters or features, track them between frames, associate detections with catalog entries, reject false matches, and feed accepted bearings into the filter. Out of scope for this paper; deferred to the future-work agenda in Section 11 along with lighting models and surface-map uncertainty.

The L1 and L2 sweeps each run three configurations on identical seeds at $n = 1000$ trials per configuration: Moon-center only, landmark-only with the Moon-center disabled, and Moon-plus-landmarks. Consistency comparison across configurations uses the pooled `nis_mean_all` column rather than the Moon-only `nis_mean`, because two of the three configurations include landmark accepts. The L1 result is summarized in Table 6 and Figure 15. Median terminal miss drops from 60.8 km (Moon-only) to 25.8 km (landmarks only) and 23.2 km (Moon+landmarks); 95th-percentile miss drops from 225.2 km to 71.1 km and

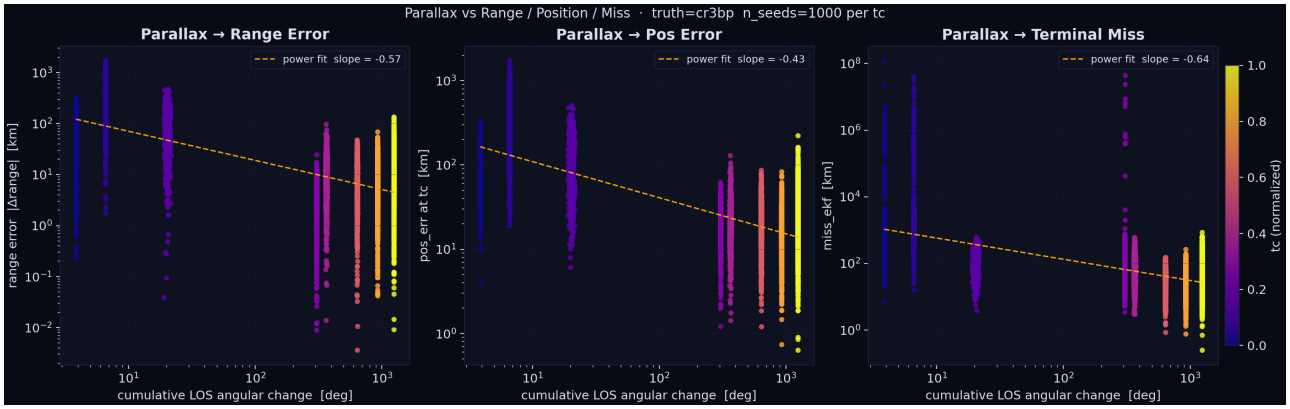

Figure 14: Cumulative LOS angular change at the burn epoch versus three error metrics, $n = 1000$ trials per correction-time bin. Each panel includes a power-law fit; the negative slope indicates that range information accumulates through bearing geometry alone. The horizontal axis is cumulative parallax (sum of step-to-step LOS deltas), which is the geometrically appropriate measure on multi-revolution arcs.

Table 6: L1 synthetic-landmark configurations, $n = 1000$ trials each, identical seeds across configurations. Pooled NIS is computed over Moon-center plus accepted landmark updates.

Configuration	miss med. [km]	miss p95 [km]	pos err at t_c [km]	NEES med.	NIS med. (pooled)
moon_only	60.80	225.16	19.06	5.78	1.94
landmarks_only	25.81	71.11	7.56	4.60	1.98
moon_plus_landmarks	23.19	69.00	7.05	4.69	1.98

Table 7: L2 catalog-backed landmark configurations, $n = 1000$ trials each. The six landmarks are Tycho, Copernicus, Aristarchus, Plato, Kepler, and Grimaldi, with selenographic coordinates converted to Moon-fixed Cartesian and identity assumed known. Same seeds as Table 6, so per-trial deltas across the L1/L2 layouts are meaningful.

Configuration	miss med. [km]	miss p95 [km]	pos err at t_c [km]	NEES med.	NIS med. (pooled)
moon_only	60.80	225.16	19.06	5.78	1.94
landmarks_only	26.57	72.89	7.71	4.59	1.98
moon_plus_landmarks	24.73	71.52	7.27	4.73	1.98

69.0 km. The improvement is therefore roughly a factor of 2.6 in median miss and 3.2 in $p95$, and the fact that landmark-only nearly matches the combined configuration indicates that the gain comes from spatial diversity of the bearing rays rather than simply more updates. The L2 catalog-backed result is reported in Table 7. Despite the catalog craters being clustered on the lunar nearside rather than spaced symmetrically on the body axes, the L2 numbers are within 6% of the L1 numbers across all three configurations (median miss 23.2/24.7 km L1/L2 in moon_plus_landmarks; $p95$ 69.0/71.5 km), showing that the conclusion is not an artifact of the idealized symmetric L1 layout. The reportable claim across both levels is that adding synthetic known landmarks substantially improves the simulated bearing geometry; the experiment does not include real crater identification, lighting and surface-map uncertainty, or false-match logic, and is therefore not a flight-relevant landmark-tracking demonstration.

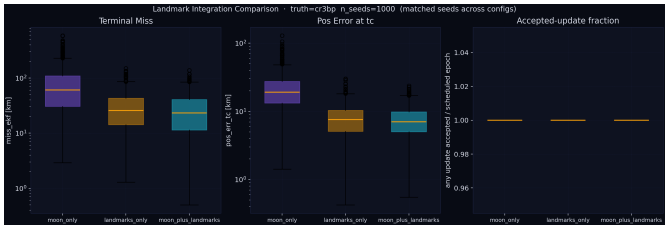

Figure 15: L1 synthetic-landmark integration comparison, $n = 1000$ trials per configuration, matched seeds across configurations. Boxes show terminal miss, position error at t_c, and the accepted-update fraction (any source) per scheduled epoch. Landmarks substantially improve the simulated bearing geometry; this experiment does not include real surface-feature identification.

13.5 Filter-Covariance Geometry at Burn

The estimation-to-guidance coupling argument made in Section 13 is summarized at the burn epoch by plotting the IEKF position covariance directly. Figure 16 shows the 1σ and 3σ covariance ellipses for the position state at t_c in three orthogonal planes for a representative seed, with the realized truth residual overlaid as a sanity check. The ellipses are not isotropic: the position uncertainty has an elongated direction that aligns with the weak observability mode reported in the bearing-only observability section. Whether that elongated direction matters for guidance depends on the alignment with the burn sensitivity, which is precisely the coupling map discussed in the guidance-coupling section. This figure makes the visual link between estimator uncertainty geometry and burn-cost sensitivity concrete.

13.6 Initial-Covariance Robustness

The headline study uses $\mathbf{P}_0 = \mathrm{diag}(10^{-6}\mathbf{I}_3, 10^{-7}\mathbf{I}_3)$. To check that the success rate is not an artifact of an opti-

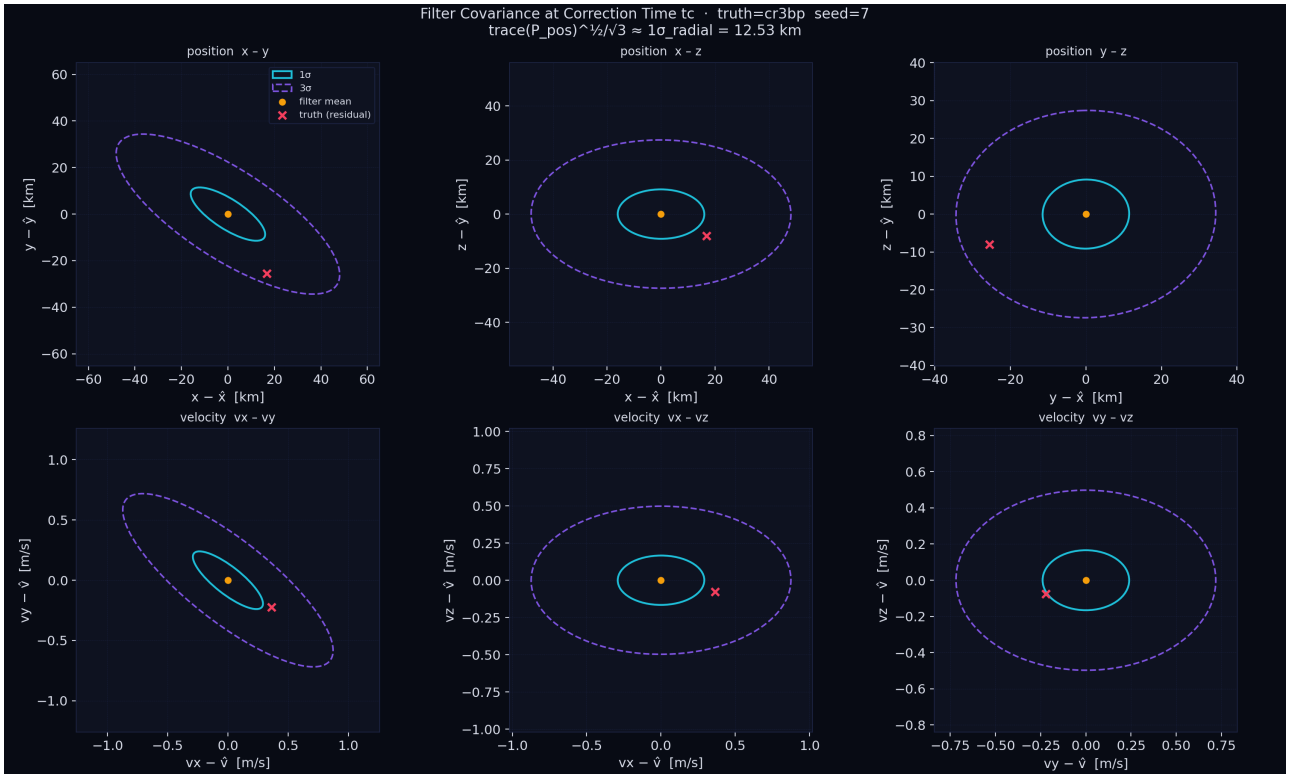

Figure 16: IEKF 1σ and 3σ covariance ellipses at the correction epoch, position panels in x–y, x–z, y–z and the corresponding velocity panels, for a representative seed. The realized truth residual is plotted as a red cross. The non-isotropic shape is a property of the bearing-only observability geometry; whether the long axis is mission-critical depends on its projection through the guidance Jacobian.

Table 8: Initial-covariance scaling sweep, $n = 1000$ trials per $\mathbf{P}_0$ scale. Median miss inflates by a factor of about 1.9 across two decades in initial covariance while filter consistency remains nominal.

$\mathbf{P}_0$ scale	n	miss median [km]	miss p95 [km]	NEES med.	NIS med.
×1	1000	60.80	225.16	5.78	1.94
×3	1000	66.74	218.12	5.18	1.94
×10	1000	81.62	262.83	5.21	1.94
×30	1000	99.74	303.94	5.59	1.94
×100	1000	114.81	356.18	6.33	1.95

mistically tight initialization, the diagonal of $\mathbf{P}_0$ is scaled by factors $\{1, 3, 10, 30, 100\}$ at $n = 1000$ trials each. Table 8 reports median and 95th-percentile terminal miss together with median NIS and NEES. Median miss inflates from 60.8 km at the baseline scale to 114.8 km at $\mathbf{P}_0 \times 100$ (a less than 2× degradation across two orders of magnitude in initial covariance), and NIS stays essentially flat at the baseline value of 1.94 across all scales. The headline result is therefore not a tightness artifact: the filter recovers a comparable estimate from a 100× wider initial covariance.

13.7 Multi-Revolution Stability

The estimator is run across longer observation arcs to test whether bearing-only navigation remains stable over multiple halo periods. Setting the correction time to $t_c = 0.95\, t_f$ and sweeping $t_f \in \{6, 9, 12\}$, which corresponds to roughly 1.9, 2.9, and 3.8 halo periods near L_1, at $n = 100$ deterministic seeds per t_f. The endpoint position error remains bounded across all arcs (median final pos. error of 11, 38, and 11 km respectively), so the filter does not diverge over multiple cycles. The fraction of trial-mean NEES values inside the $\chi^2(6)$ 95% interval is 49.1%, 34.7%, and 26.8% as the arc length grows; this is a consistency degradation, not a divergence, and is consistent with mild over-confidence accumulating over extended arcs. The reportable claim is that the filter is stable in the energy sense over multi-revolution arcs but increasingly tight on its own covariance, which is a tunable defect rather than a structural failure.

13.8 Measurement-Delay Sensitivity

The Section 13-style realism extension that the verification section already flags as the most operationally dangerous case is measurement delay. Sweeping $\Delta k \in \{0, 1, 2, 5, 10, 20, 50\}$ measurement steps, with the truth-side image generated at the delayed epoch (and, under SPICE, the Moon ephemeris also fetched at the delayed epoch for geometric consistency), produces Table 9. A single-step delay ($\Delta t = 0.02$ ND, roughly 2 hours of physical time at the Earth–Moon scale) inflates the median terminal miss from 60.8 km to 918.5 km, a 15× degradation,

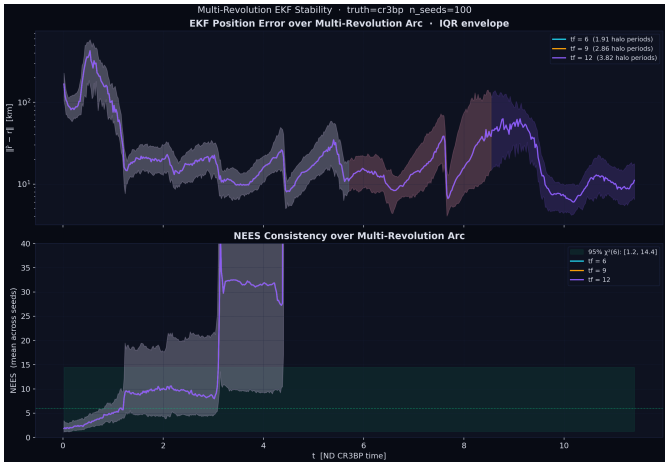

Figure 17: Multi-revolution position error and NEES envelopes (median and inter-quartile band), $n = 100$ seeds per arc length. Endpoint position error remains bounded across arcs of 1.9 to 3.8 halo periods; NEES drifts inside the diagnostic interval rather than diverging.

Table 9: Measurement-delay sweep, $n = 1000$ trials per delay step. Median miss inflates by an order of magnitude on a single delayed step while the median Moon-center NIS only moves from 1.94 to 2.04; the median NEES diverges instead. Δt is in CR3BP non-dimensional time units (one ND time ≈ 4.34 days physical time at the Earth–Moon scale).

Δk [steps]	Δt [ND]	miss median [km]	miss p95 [km]	NIS med.	NEES med.
0	0.000	60.80	225.16	1.94	5.78×10^{0}
1	0.020	918.50	1945.55	2.04	9.89×10^{7}
2	0.040	983.64	2671.89	2.37	1.00×10^{9}
5	0.100	1447.20	3523.00	7.59	7.16×10^{9}
10	0.200	11655.21	15609.92	61.29	1.47×10^{10}
20	0.400	901428.47	36439165.25	286.75	7.40×10^{10}
50	1.000	694992.40	27490449.38	160.74	2.53×10^{10}

while the median NIS climbs only modestly from 1.94 to 2.04 and remains within the diagnostic $\chi^2(2)$ 95% band. The median NEES, in contrast, balloons by eight orders of magnitude in a single step. This is the strongest evidence in the present pipeline that NIS in isolation is not sufficient as a navigation-health metric: the filter believes its measurements at the wrong time-tag are still consistent residuals, while the state-space reality has already drifted out of bounds.

14 Verification and Stress Testing

14.1 Numerical Verification

Several numerical checks were used to separate navigation conclusions from implementation artifacts. The propagation path advances the nonlinear state and STM together with adaptive integration tolerances, then uses the STM to propagate covariance between measurement epochs. The IEKF update uses the Joseph covariance form and Cholesky-based innovation solves where possible. Covariance matrices are symmetrized and checked for positive definiteness after propagation and update, and Monte Carlo trials are generated from deterministic seeds so that tables and plots can be reproduced from the accompanying numerical products.

14.2 Estimator Consistency

The tangent-plane residual is two-dimensional, so per-update NIS is compared against a $\chi^2(2)$ distribution. In the final baseline, the median trial-mean NIS was 1.94 and all 1000 trial-mean NIS values fell inside the diagnostic $\chi^2(2)$ 95% interval. The six-state NEES was checked against $\chi^2(6)$; 99.4% of baseline trial-mean NEES values fell inside the diagnostic 95% interval $[1.237, 14.449]$. Individual-run diagnostics additionally include image-plane residuals, visibility flags, innovation histories, NEES/NIS histories, and position/velocity 3σ envelopes.

14.3 SPICE Point-Mass Truth Stress Test

A 1000-trial CR3BP-versus-SPICE comparison was generated as a model-mismatch stress test. The truth model in the SPICE arm is a Sun–Earth–Moon point-mass propagator seeded from JPL ephemerides; it is not a full flight-fidelity model with non-spherical gravity, solar radiation pressure, attitude dynamics, or sensor errors. The exercise is therefore best read as a SPICE point-mass truth stress test: it intentionally applies the CR3BP-tuned estimator to a higher-fidelity ephemeris-based truth to identify which metrics are sensitive to model mismatch. The same one-pixel measurement model and estimator structure are run against both CR3BP truth and SPICE truth, with CR3BP outcomes converted to kilometres for comparison. Figure 18 shows the key lesson. The median correction-time position error remains nearly the same, about 19 km in both runs, and NIS remains close to the expected $\chi^2(2)$ behavior. The terminal-miss distribution, however, shifts from a median of about 62 km under CR3BP truth to about 183 km under SPICE truth, and the NEES tail grows. Innovation consistency alone therefore does not fully validate model fidelity; state consistency and guidance performance expose errors that measurement residuals can hide.

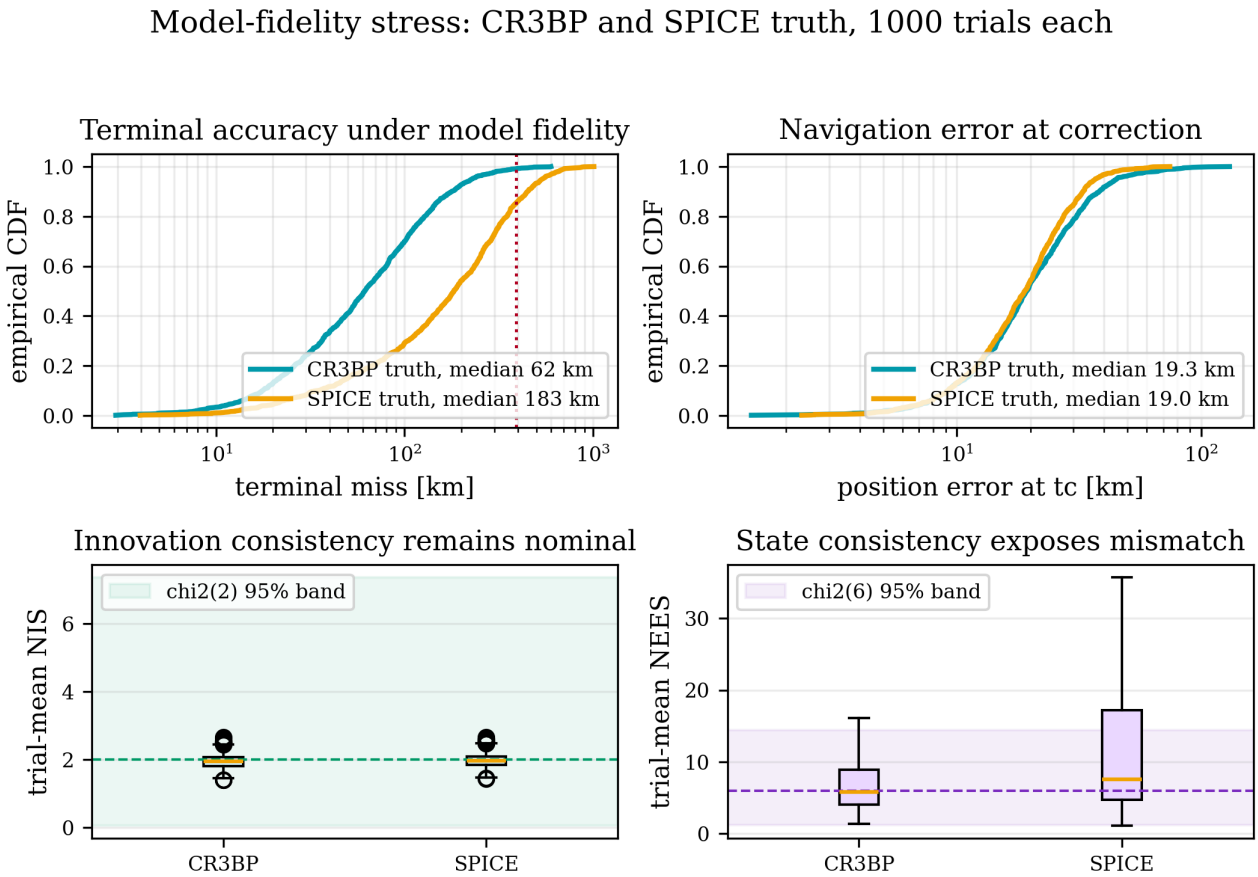

Figure 18: Model-fidelity stress summary for 1000-trial CR3BP and SPICE point-mass truth cases. CR3BP errors are converted to kilometres using the Earth–Moon length scale. NIS remains nominal while NEES and terminal miss reveal the effect of higher-fidelity ephemeris-based truth.

14.4 Fault and Timing Sensitivity

Table 10 summarizes representative diagnostic cases. The key qualitative result is that estimate-tracking and truth-tracking are nearly identical in the nominal case, which verifies that active pointing itself does not introduce a significant bias when driven by a good estimate. Random dropout and pixel outlier scenarios remain stable in the tested settings. A one-step measurement delay, however, fails: NIS can still look acceptable while state error grows, which is a warning that innovation consistency alone is not sufficient for validating time alignment.

15 Discussion

The strongest conclusion is that measurement availability is just as important as measurement noise. The bearing-only IEKF can produce accurate estimates when the Moon remains in the field of view, but fixed pointing can starve the filter of updates. Active pointing changes the problem: it converts a fragile passive camera into a closed-loop measurement source. In this sense, pointing is not merely an attitude-control detail; it is part of the observability strategy.

Table 10: Selected diagnostic-suite outcomes. Final errors are nondimensional.

Case	Valid rate	NIS mean	Final $\lvert\mathbf{r}\rvert$ error
Fixed pointing	0.257	440.6	4.86×10^{-1}
Truth tracking	1.000	1.754	6.14×10^{-5}
Estimate tracking	1.000	1.754	6.14×10^{-5}
Dropout	0.960	1.792	5.26×10^{-5}
Pixel outliers	1.000	1.716	7.32×10^{-5}
One-step delay	0.977	1.718	2.96×10^{-3}

The second conclusion is that a good-looking trajectory estimate is not enough. Early versions of the filter produced small errors while producing covariances that were too tight. The q_a sweep and NEES checks distinguish accuracy from consistency. This matters because a guidance system consumes not only the state estimate, but often also the uncertainty used for gating, planning, and fault response.

The third conclusion is that observability must be interpreted through the guidance problem. The Gramian identifies weak directions, but the coupling map shows which of those directions actually damage the burn. The analysis does not stop at whether the filter settles; it asks whether the recovered state projects into acceptable ter-

minal miss and burn deviation.

Preliminary feature tracking and known-pose landmark reconstruction were implemented as a feasibility step. In Section 13.4 two levels of landmark-bearing fusion are folded into the estimator and compared against Moon-center only on identical seeds: L1 with six synthetic Moon-fixed offsets along the body axes, and L2 with selenographic latitude/longitude coordinates for six well-known nearside craters with identity assumed known. Both levels run $n = 1000$ trials per configuration. The result is a measurement-geometry comparison, not a flight-relevant landmark-tracking demonstration; real crater detection from images, lighting and surface-map uncertainty, and false-match logic are deferred to L3 future work. The reportable claim is that adding known landmarks substantially improves the simulated bearing geometry, and that this conclusion holds under both an idealized symmetric L1 layout and a realistic asymmetric L2 catalog distribution.

The realism sweeps make the dependence of the headline result on its assumptions explicit. Tightening the success threshold to 10^{-4} LU $\approx$ 39 km drops the pass rate from 99% to 33% on CR3BP truth and from 85% to 11% on SPICE point-mass truth, so the baseline architecture meets a generous tolerance comfortably but is not a precision-arrival tool out of the box. Adding zero-mean attitude noise of up to 0.03 degrees per step degrades miss only marginally (within five percent of the no-noise baseline); the median miss inflates by about 43% at 0.1 degrees, then climbs steeply to roughly 1000 km at 1 degree because the un-modeled bearing residual exceeds the pixel-noise contribution and the filter loses statistical consistency. The cumulative-parallax versus range-error fit recovers a negative-exponent power law (slope ≈ -0.57 on the production data), which is the bearing-only physics-backed reason the filter ever resolves range, while making clear that range information is bounded by both an insufficient-arc floor at small t_c and a noise floor at long arc. The synthetic-landmark experiment improves median terminal miss by a factor of about 2.6 relative to Moon-center only, and the landmark-only configuration nearly matches the combined Moon-plus-landmarks configuration, suggesting that bearing-ray spatial diversity is the dominant effect rather than the number of accepted updates. Together, these tests support the headline claim while delineating where it stops being valid.

15.1 Modeling Assumptions and Limitations

Several simplifying assumptions are important to make explicit. The main estimator and Monte Carlo studies use CR3BP dynamics; the SPICE arm uses a Sun–Earth–Moon point-mass propagator seeded from JPL ephemerides, which is best characterized as a point-mass truth stress test rather than a full flight-fidelity simulation. Higher-order non-spherical gravity, solar radiation pressure, and propellant-related disturbances are not modeled. The target-body ephemeris is assumed known, light-time and aberration effects are neglected, and the image-to-bearing conversion assumes perfect camera calibration apart from injected pixel noise. The pointing law is idealized in baseline runs and explicitly stress-tested in the realism sweeps via zero-mean attitude noise, deterministic pointing bias, and pointing-lag steps; actuator limits, slew dynamics, and closed-loop attitude estimation remain out of scope. The synthetic-landmark experiment of Section 13.4 demonstrates a measurement-geometry effect, not real surface-feature identification: limb shape, illumination bias, star-background confusion, false-match logic, and surface-map uncertainty are not modeled.

The single-target bearing-only architecture has understandable limits. It performs well for the nominal one-pixel baseline and a tuned covariance model, but high pixel noise, measurement delay, large pointing bias, early correction time, very poor initialization, or model mismatch can degrade the correction result. Future improvements should prioritize real landmark identification with map and lighting uncertainty, multi-body bearings, explicit time-tag estimation in the filter, optical calibration error estimation, and higher-fidelity dynamics that include non-spherical gravity and solar radiation pressure.

The main threat to validity is that the strongest Monte Carlo results are obtained in a CR3BP truth environment with idealized camera calibration and one-pixel measurement noise. The SPICE point-mass truth comparison partially addresses dynamics mismatch, the realism sweeps in Section 13 address pointing and tolerance, and the measurement-delay study addresses time-tag sensitivity, but a flight-relevant assessment would require optical calibration errors, illumination-dependent center bias, full ephemeris/perturbation modeling, and surface-feature identification to be modeled or estimated.

16 Conclusion

This study numerically demonstrates that a single Moon-center bearing stream can support a midcourse correction in the tested cislunar transfer when target visibility is actively maintained. The baseline achieved 99.1% success over 1000 Monte Carlo trials for a 10^{-3} LU terminal-miss tolerance, with median miss 1.81×10^{-4} LU and median relative ΔV inflation 1.19% relative to the perfect-information burn. Filter consistency was also credible: median trial-mean NIS was 1.94, median trial-mean NEES was 4.34, and 99.4% of trial-mean NEES values were inside the diagnostic $\chi^2(6)$ 95% interval.

The technical result is not simply that an IEKF settles. The result is that bearing-only navigation must be evaluated as a coupled sensing-estimation-guidance problem. Repeated angular measurements and CR3BP dynamics can recover useful range and velocity information, but the weakest estimation directions are not automatically the most mission-critical directions; they become critical when they align with the burn sensitivity of the trans-

Table 11: Scope of validity for the main results.

Assumption	Included?	Stress-tested?	Future work
Pixel noise	Yes, Gaussian baseline	Yes, high-noise Monte Carlo and noise sweep	Higher-fidelity image and detection model
Active pointing	Ideal estimate-driven pointing	Yes: fixed versus active comparison plus zero-mean attitude noise sweep, deterministic pointing bias, and pointing-lag steps	Slew limits, actuator dynamics, closed-loop attitude estimation
Initial covariance	$\mathbf{P}_0$ fixed at the values in Section 13	Yes: $\mathbf{P}_0$-scale sweep over $\{1, 3, 10, 30, 100\}$	Adaptive initialization, recovery from very poor initial covariance
Tolerance threshold	10^{-3} LU $\approx$ 390 km success threshold	Yes: success-rate-versus-tolerance sweep down to 1 km	Mission-specific accuracy requirement
Timing error	Mostly ideal synchronized measurements	Yes: measurement-delay sweep with geometrically consistent SPICE delay	Continuous timing bias, jitter, and on-board time-tag estimation
Dynamics mismatch	CR3BP estimator with SPICE point-mass truth stress case	Yes: $n = 1000$ CR3BP/SPICE point-mass comparison; multi-revolution CR3BP arc	Retuned high-fidelity filter and full ephemeris/perturbation model
Calibration	Perfect intrinsics except injected pixel noise	No	Optical calibration uncertainty and bias estimation
Landmarks	L1 synthetic-bearing and L2 catalog-backed cases (identity assumed known); no L3 image-recognition pipeline	Yes: L1 synthetic six-landmark and L2 six-crater catalog configurations versus Moon-only on identical seeds, $n = 1000$ each	L3: image-based detection, frame-to-frame tracking, catalog association, false-match rejection, lighting and surface-map uncertainty
Illumination and star confusion	Excluded	No	Limb/photometric center model and star-field rejection

fer. NIS alone is therefore too narrow a success criterion: NEES, terminal miss, and burn deviation are needed to expose overconfidence and model mismatch.

Flight relevance requires additional observables and error sources beyond this minimal-information baseline. This preprint presents the Moon-center bearing-only baseline together with reviewer-driven realism sweeps that stress-test pointing, terminal tolerance, parallax-based range observability, initial covariance, multi-revolution stability, and a synthetic-landmark measurement-geometry variant on top of the existing CR3BP/SPICE point-mass truth comparison. Real surface-feature identification with map and lighting uncertainty, multi-body bearings, optical calibration error estimation, on-board time-tag estimation, and higher-fidelity dynamics that include non-spherical gravity and solar radiation pressure remain the natural next steps.

Code Availability

The source code and numerical products required to reproduce the figures and tables are available in the accompanying repository. The main experiments include the baseline Monte Carlo study, process-noise sweep, active-pointing comparison, sensitivity studies, and model-mismatch stress test.

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